



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1-AT3-TECHNICAL PRESENTATION

CSA0504 : Database Management Systems

PRESENTED BY :

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Question 1: ER Diagram for an Online Library System



INTRODUCTION

- ER Diagram stands for **Entity Relationship Diagram**.
- It represents the structure of a database visually.
- It shows:
 - 1)Entities (objects)
 - 2)Attributes (properties)
 - Relationships (connections)
- Used to design and manage library databases efficiently.
- **Key Point:** Helps in organizing books, members, and borrowing records.

MAIN ENTITIES AND ATTRIBUTES

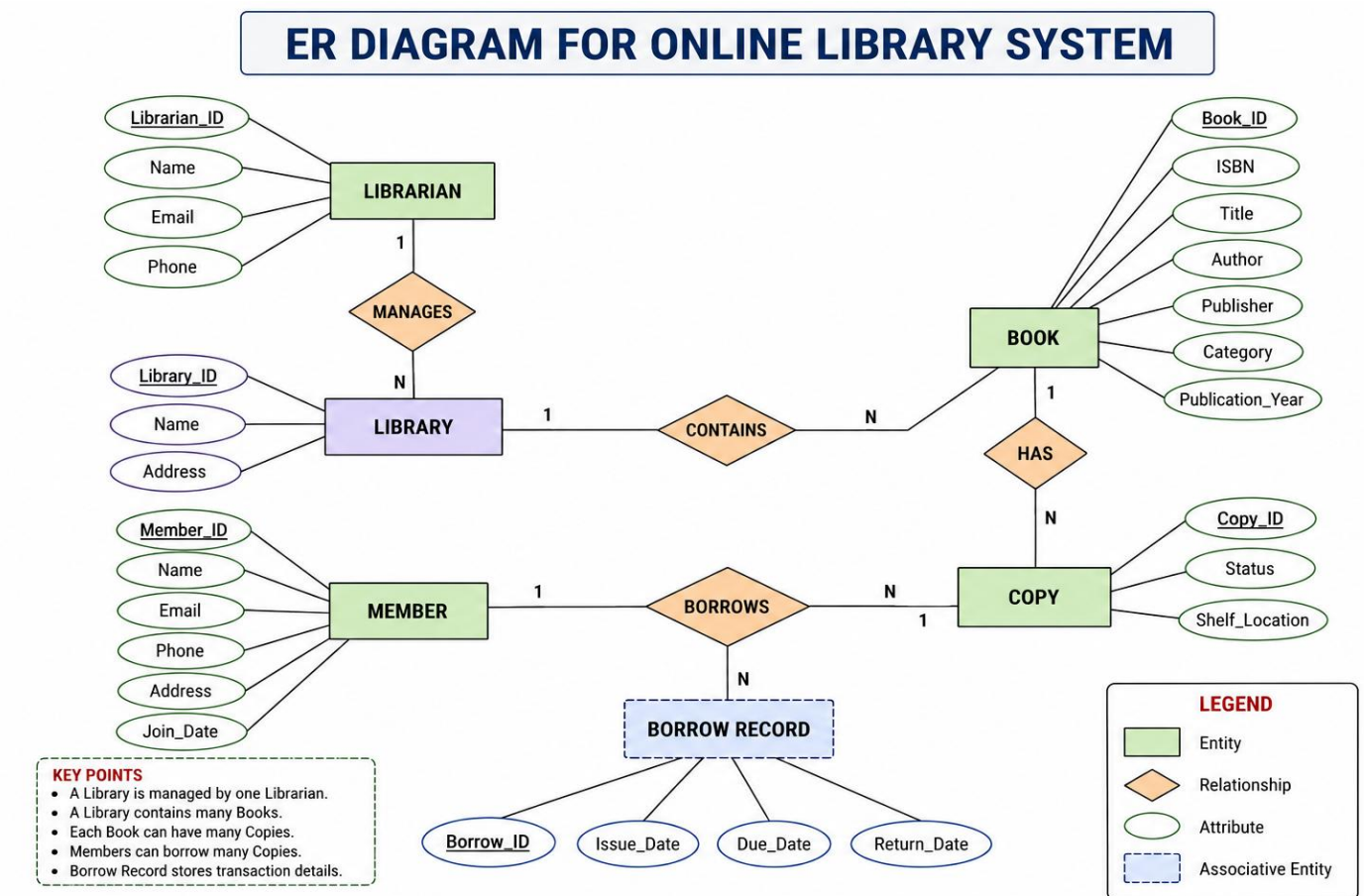
Entity	Important Attributes
Library	Library_ID, Name, Location
Book	Book_ID, Title, Author, ISBN
Member	Member_ID, Name, Email, Phone
Librarian	Librarian_ID, Name
Borrow Record	Borrow_ID, Issue_Date, Return_Date

Important Entities:

1)Book 2)Member 3)Librarian 4)Borrow Record 5)Library

RELATIONSHIPS IN THE ER DIAGRAM

- A **Library** contains many **Books** and a **Member** can borrow multiple **Books**.
- A **Librarian** manages books and members and **Borrow Record** connects Members and Books.
- Relationships:
 - Library → Contains → Books
 - Member → Borrows → Books
 - Librarian → Manages → Library
 - Borrow Record → Tracks → Issue & Return dates .



Conclusion : The ER Diagram provides a clear representation of the Online Library System, making database design simple, organized, and efficient.



Question 2: EER Model – University Case Study



INTRODUCTION

- EER stands for **Enhanced Entity Relationship Model**.
- It is an extension of the ER Model with advanced features.
- EER includes:
 - 1)Specialization
 - 2)Generalization
 - 3)Inheritance
 - 4)Aggregation
- It is used to model complex database systems such as universities.
- **Key Point:** The EER model provides a detailed and flexible representation of real-world scenarios.

University Case Study

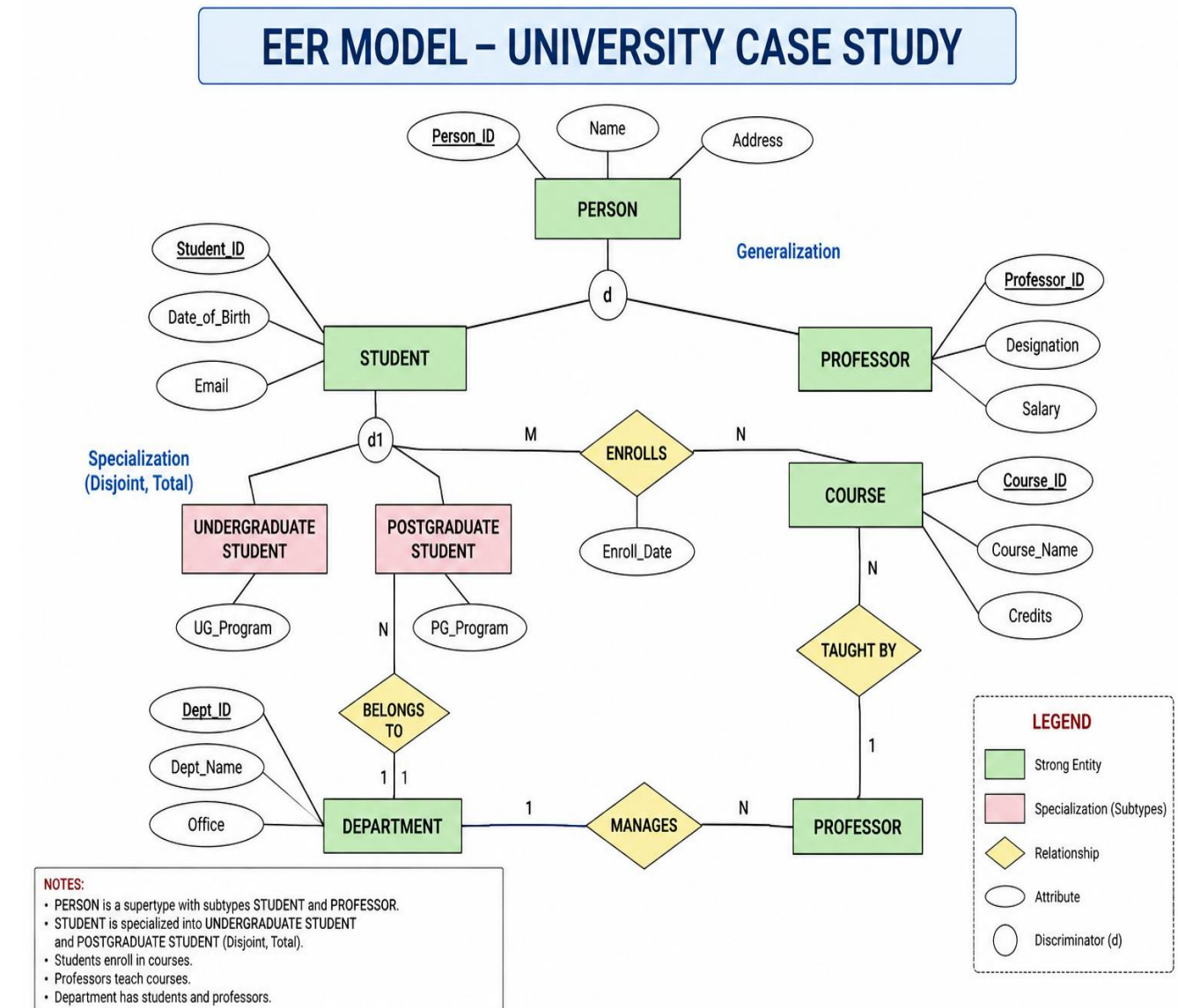
Entity	Attributes
Student	Student_ID, Name, Department
Professor	Professor_ID, Name, Designation
Course	Course_ID, Course_Name, Credits
Department	Department_ID, Department_Name
Classroom	Room_No, Building_Name

Important Entities:

1)Student 2)Professor 3)Course 4)Department 5)Classroom

Relationships and EER Concepts

- A **Student** enrolls in multiple **Courses**.
- A **Professor** teaches one or more **Courses**.
- A **Department** contains Students and Professors.
- Specialization Example:
 - Student → Undergraduate Student
 - Student → Postgraduate Student
- Generalization Example:
 - Employee → Professor, Administrative Staff



Conclusion : The EER Model helps represent complex relationships and inheritance in a university database system, making it more efficient and easier to manage.



Question 3: Data Independence in DBMS



INTRODUCTION

- Data Independence is the ability to modify the database schema without affecting application programs.
- It is an important feature of DBMS.
- It helps in maintaining and updating databases efficiently.
- It provides flexibility and reduces maintenance costs.
- **Key Point:** Changes in one level of the database should not impact other levels.

Types of Data Independence

Physical Data Independence	Logical Data Independence
Changes in physical storage do not affect the logical schema.	Changes in the logical schema do not affect external views.
<u>Examples:</u> Changing file organization. Adding indexes. Modifying storage devices.	<u>Examples:</u> Adding new attributes. Creating new tables. Modifying relationships.



Engineer to Excel



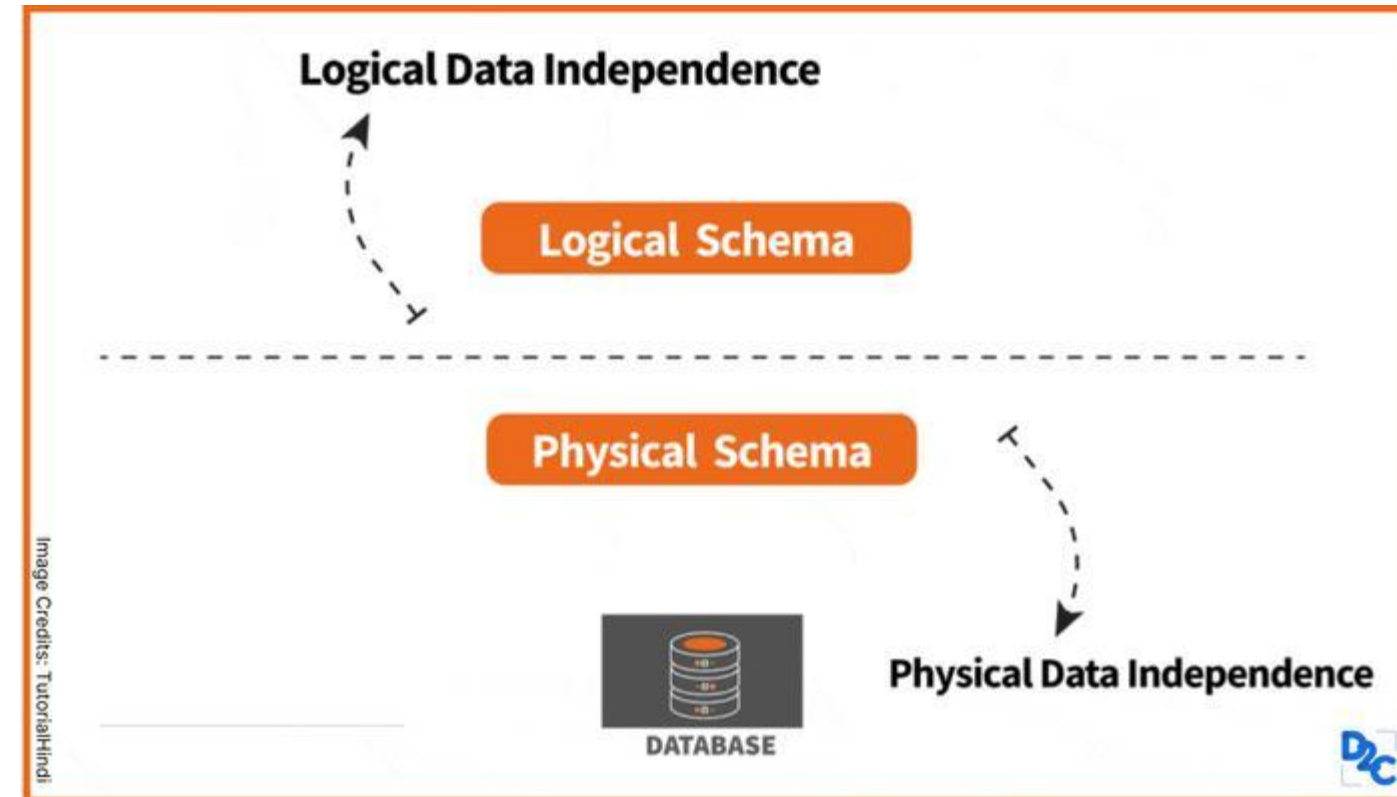
Benefits and Conclusion

Benefits:

- Easy database maintenance.
- Improved flexibility.
- Better data security.
- Reduced application changes.
- Supports database scalability.

Conclusion:

Data Independence is a fundamental concept in DBMS that ensures changes at one level of the database architecture do not affect other levels, making systems more robust and easier to manage.

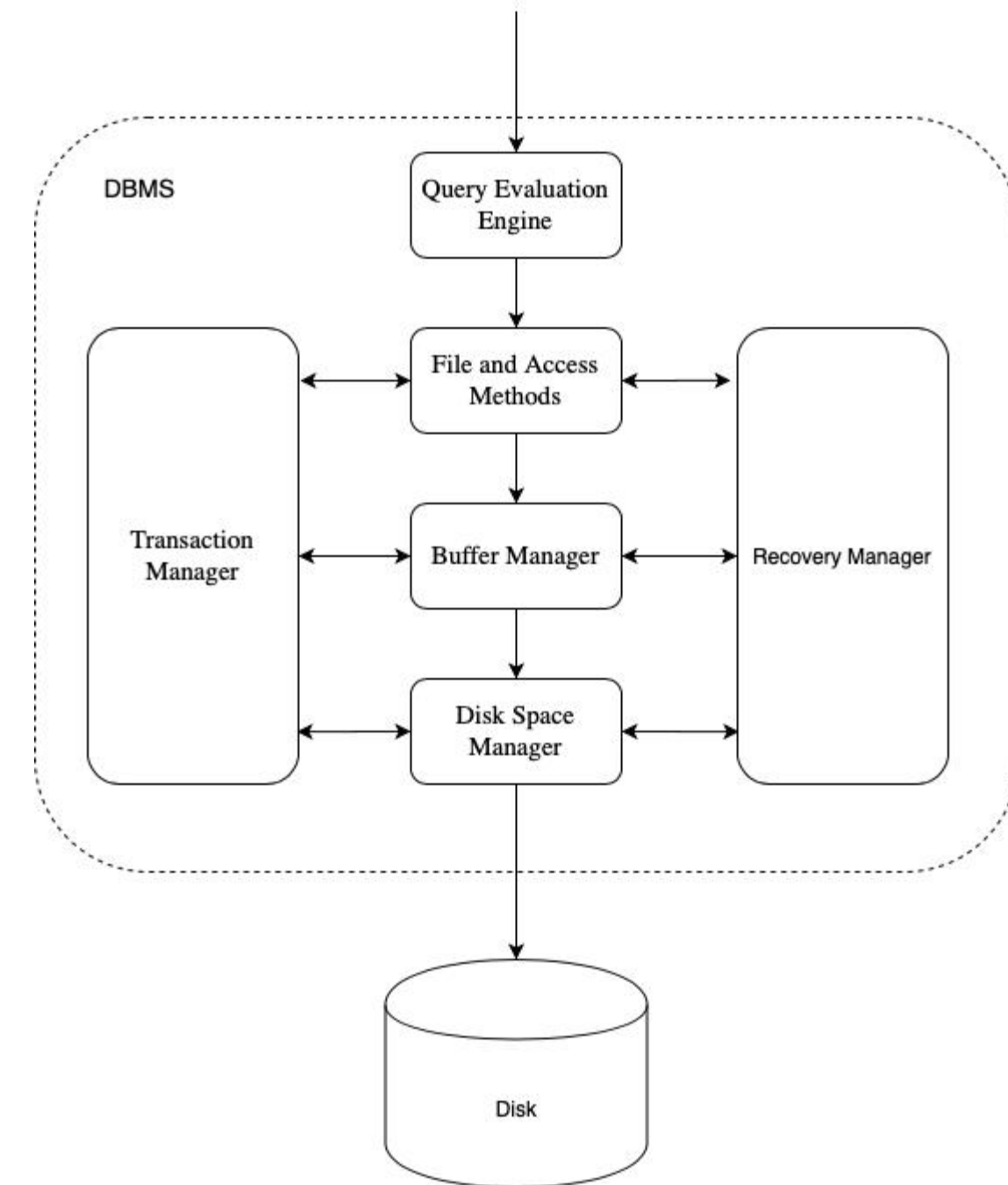




Question 4: Software Components of DBMS

INTRODUCTION

- DBMS (Database Management System) consists of several software components that work together to manage databases.
- These components ensure efficient storage, retrieval, and security of data.
- They help users interact with the database smoothly.
- **Key Point:** Software components are the building blocks of a DBMS.





Main Software Components of DBMS



- 1.Query Processor** : Interprets and executes SQL queries.
- 2.Storage Manager** : Manages data storage and retrieval.
- 3.Transaction Manager** : Ensures data consistency and handles transactions.
- 4.Authorization & Integrity Manager** :Controls user access and maintains data integrity.
- 5.Buffer Manager** : Manages data transfer between memory and disk.
- 6.File Manager** : Organizes files on secondary storage.



Benefits & Conclusion



Benefits:

Efficient data management.

Improved security.

Faster query processing.

Better transaction handling.

Reliable data storage.

Conclusion:

The software components of DBMS work together to provide secure, efficient, and reliable database management, ensuring smooth operation of modern database systems.



Question 5: Strong and Weak Entities



Introduction to Strong and Weak Entities

- Entities are objects or concepts represented in an ER Diagram.
- There are two types of entities in DBMS:
 - 1) Strong Entity
 - 2) Weak Entity
- They help in designing relationships within a database.
- **Key Point:** A weak entity depends on a strong entity for its existence.

Strong Entity vs Weak Entity

Strong Entity	Weak Entity
Has its own primary key.	Does not have a primary key.
Exists independently.	Depends on a strong entity.
Represented by a single rectangle.	Represented by a double rectangle.
Example: Employee, Student	Example: Dependent, Order Item

Examples:

- Strong Entity: Student (Student_ID)
- Weak Entity: Dependent (Dependent_Name)

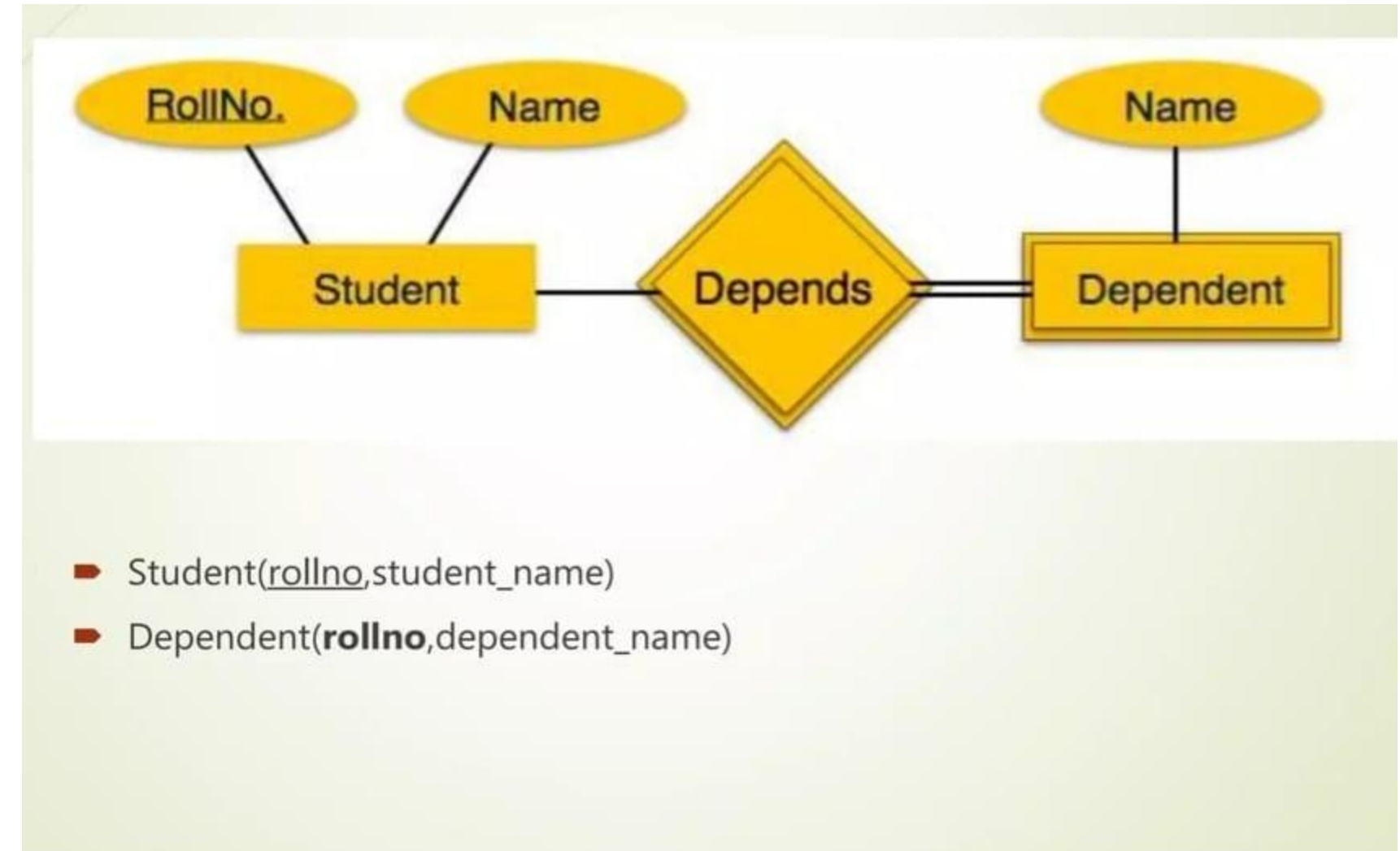
Benefits and Conclusion

Benefits:

- Helps model real-world relationships.
- Improves database design.
- Ensures data consistency.
- Simplifies entity identification.

Conclusion:

Strong and Weak Entities are fundamental concepts in ER modeling. Strong entities can exist independently, whereas weak entities rely on strong entities for identification and existence.



*Thank
you!*